**Table 3:** Incremental discrimination under L2-penalized logistic regression, with margin analysis. ΔAUC with paired-bootstrap 95% and 90% intervals (2,000 resamples). The last three columns state whether the 90% interval lies wholly within $\pm\delta$ (equivalence by two one-sided tests at $\alpha = 0.05$) for $\delta$ = 0.01, 0.015 and 0.02. The margin analysis, M4 and the M0+ − M0 contrast were added at peer review (post hoc); the M4 − M0+ lower 90% bound is −0.00997, shown rounded; the 0.02 value was declared in advance without external justification and no margin is privileged. Sensitivity analyses are in Additional file 4 (Table S8) and every pairwise contrast under every algorithm in Table S6.

| | | | | | Within $\pm\delta$ | | |
|---|---|---|---|---|---|---|---|
| Contrast | | ΔAUC | 95% CI | 90% CI | 0.01 | 0.015 | 0.02 |
| Metals added to demographics | M1 − M0 | −0.0015 | [−0.0052, 0.0023] | [−0.0047, 0.0018] | yes | yes | yes |
| Nutritional and metabolic biomarkers added to demographics | M2 − M0 | 0.0011 | [−0.0067, 0.0100] | [−0.0055, 0.0087] | yes | yes | yes |
| Both biomarker blocks added to demographics (primary) | M3 − M0 | −0.0019 | [−0.0108, 0.0071] | [−0.0094, 0.0056] | yes | yes | yes |
| Clinical covariates added to demographics (post hoc) | M0+ − M0 | 0.0139 | [0.0026, 0.0255] | [0.0043, 0.0232] | no | no | no |
| Both biomarker blocks added to the clinical baseline (post hoc) | M4 − M0+ | −0.0037 | [−0.0112, 0.0029] | [−0.0100, 0.0019] | yes | yes | yes |
| Clinical baseline against biomarker model (substitutes) | M0+ − M3 | 0.0158 | [0.0033, 0.0295] | [0.0055, 0.0270] | no | no | no |